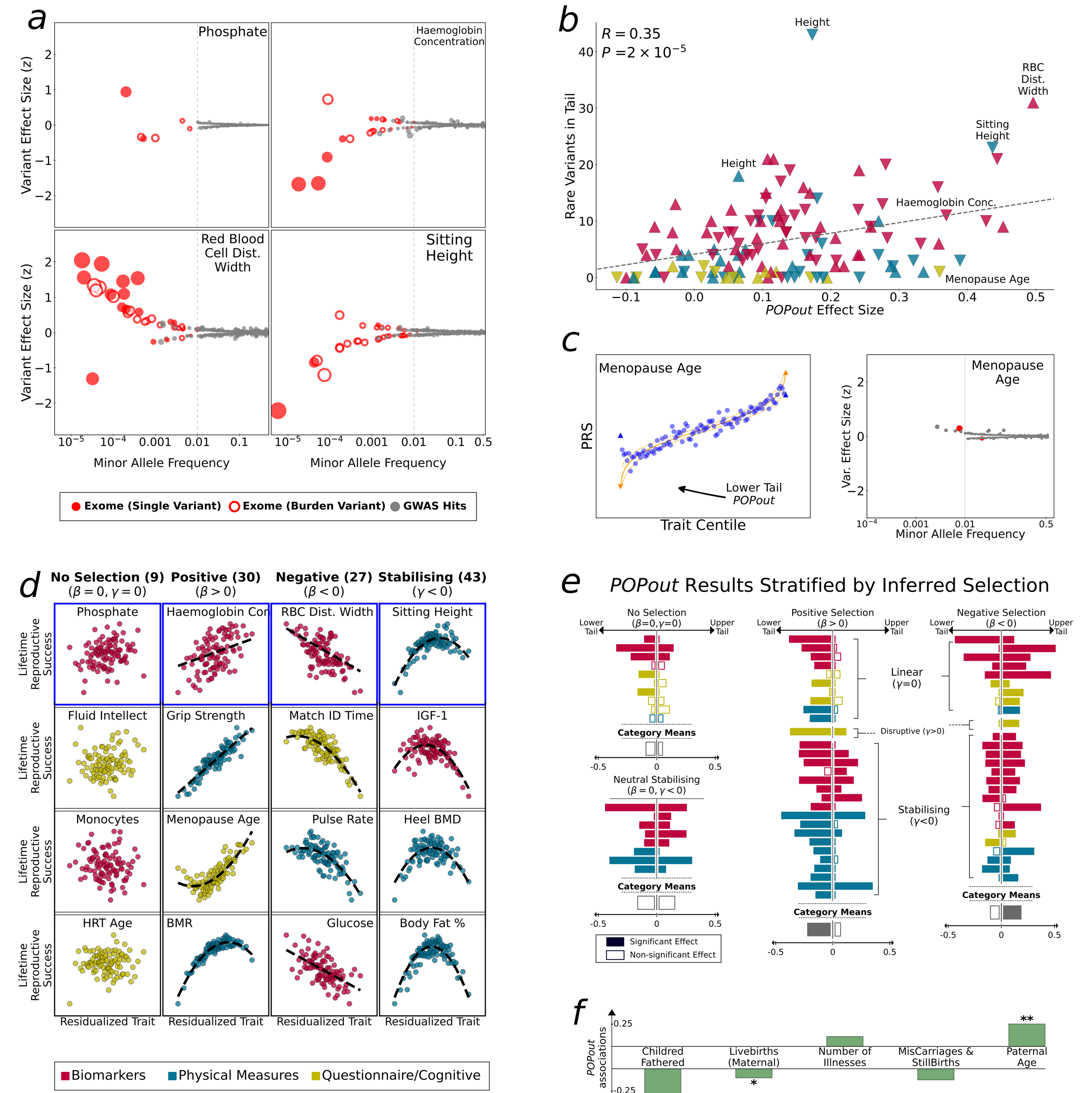
a
Phosphate
Haemoglobin Concentration
Red Blood Cell Dist. Width
Sitting Height
Variant Effect Size (z)
Minor Allele Frequency
Exome (Single Variant)
Exome (Burden Variant)
GWAS Hits
b
R = 0.35
P = 2 × 10^−5
Height
RBC Dist. Width
Sitting Height
Height
Haemoglobin Conc.
Menopause Age
Rare Variants in Tail
POPout Effect Size
c
Menopause Age
PRS
Lower Tail POPout
Trait Centile
Menopause Age
Var. Effect Size (z)
Minor Allele Frequency
d
No Selection (9) (β = 0, γ = 0)
Positive (30) (β > 0)
Negative (27) (β < 0)
Stabilising (43) (γ < 0)
Phosphate
Haemoglobin Con
RBC Dist. Width
Sitting Height
Fluid Intellect
Grip Strength
Match ID Time
IGF-1
Monocytes
Menopause Age
Pulse Rate
Heel BMD
HRT Age
BMR
Glucose
Body Fat %
Lifetime Reproductive Success
Residualized Trait
Biomarkers
Physical Measures
Questionnaire/Cognitive
e
POPout Results Stratified by Inferred Selection
No Selection (β=0,γ=0)
Positive Selection (β > 0)
Negative Selection (β < 0)
Lower Tail
Upper Tail
Category Means
Neutral Stabilising (β = 0, γ < 0)
Linear (γ=0)
Disruptive (γ>0)
Stabilising (γ<0)
Significant Effect
Non-significant Effect
f
POPout associations
Childred Fathered
Livebirths (Maternal)
Number of Illnesses
MisCarriages & StillBirths
Paternal Age
*P < 0.05, **P < 0.01, ***P < 0.001